Section 31 A/D Converter (ADCA)

This section contains a generic description of the A/D Converter (ADCA).

The first part of this section describes all RH850/F1K specific properties, such as the number of units, register base addresses, etc. The remainder of the section describes the functions and registers of the ADCA.

31.1 Features of RH850/F1K ADCA

31.1.1 Number of Units and Channels

This microcontroller has the following number of ADCA units.

Table 31.1 Number of Units

Product Name	RH850/F1K 100 pins	RH850/F1K 144 pins	RH850/F1K 176 pins
Number of Units	1	2	2
Name	ADCAn (n = 0)	ADCAn (n = 0, 1)	ADCAn (n = 0, 1)

An ADCAn unit has the same number of physical channels as the number of A/D input pins and the same number of virtual channels as the number of addresses where the results of A/D conversion will be stored. The numbers of channels on individual products are as listed below.

Table 31.2 Unit Configurations and Physical Channels

Unit Name (Number of Channels) ADCAn		RH850/F1K 100 pins	RH850/F1K 144 pins	RH850/F1K 176 pins
ADCA0	12 bit pin for conversion*1	16	16	16
	10 bit pin for conversion*2	20	20	20
ADCA1	12 bit pin for conversion*1	—	8	16
	10 bit pin for conversion*2	—	4	8

Note 1. When 10-bit mode is selected, this pin can be used for 10-bit conversion.

Note 2. When 12-bit mode is selected but a pin is for 10-bit conversion, the 2 low-order bits of the result of conversion must be masked before use.

Table 31.3 Unit Configurations and Virtual Channels

Unit Name (Number of Channels) ADCAn	RH850/F1K 100 pins	RH850/F1K 144 pins	RH850/F1K 176 pins
ADCA0	50	50	50
ADCA1	—	36	36

Table 31.4 Indices

Index	Description
n	Throughout this section, the individual ADCA units are identified by the index “n” (n = 0, 1); for example, ADCAnPWDVCR indicates the PWM-Diag virtual channel register.
m	Throughout this section, the individual physical channels (channels in the unit) of ADCAn are identified by the index “m”; for example, ANInm.
j	Throughout this section, the individual virtual channels of ADCAn are identified by the index “j”; for example, ADCAnVCRj indicates the virtual channel register.
x	Throughout this section, the individual scan groups (SG) of ADCAn are identified by the index “x” (x = 1 to 3); for example, ADCAnSGSTCRx indicates the scan group x start control register.
k	Throughout this section, the individual physical channel numbers for T&H are identified by the index “k” (k = 0 to 5); for example, THkE is T&H enable bit of the T&H enable register (ADCAnTHER).

The following table shows values indicated by the indices of each product.

Table 31.5 Indices of Products

Indices of Each Product		
RH850/F1K 100 pins	**RH850/F1K 144 pins**	**RH850/F1K 176 pins**
m = 0 to 35 (ADCA0)	m = 0 to 35 (ADCA0) m = 0 to 7 (ADCA1) 16 to 19 (ADCA1)	m = 0 to 35 (ADCA0) m = 0 to 23 (ADCA1)
j = 00 to 49 (ADCA0)	j = 00 to 49 (ADCA0) j = 00 to 35 (ADCA1)	j = 00 to 49 (ADCA0) j = 00 to 35 (ADCA1)
x = 1 to 3 (ADCA0)	x = 1 to 3 (ADCA0) x = 1 to 3 (ADCA1)	x = 1 to 3 (ADCA0) x = 1 to 3 (ADCA1)
k = 0 to 5 (ADCA0)	k = 0 to 5 (ADCA0)	k = 0 to 5 (ADCA0)

31.1.2 Register Base Address

ADCAn base addresses are listed in the following table.

ADCAn register addresses are given as offsets from the base addresses.

Table 31.6 Register Base Addresses

Base Address Name	Base Address
<ADCA0_base>	FFF2 0000_H
<ADCA1_base>	FFD6 $D000_H$

31.1.3 Clock Supply

The ADCAn clock supply is shown in the following table.

Table 31.7 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name	Description
ADCA0	ADCLK	CKSCLK_AADCA	Module clock
	Register access clock	CPUCLK2	Bus clock
		CKSCLK_AADCA	
ADCA1	ADCLK	CKSCLK_IADCA	Module clock
	Register access clock	CPUCLK2	Bus clock

31.1.4 Interrupt Requests

ADCAn interrupt requests are listed in the following table.

Table 31.8 Interrupt Requests

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number	Other Trigger Signals
ADCA0				
INT_ADE	A/D error interrupt	56	—	Motor control
INT_SG1	Scan group 1 (SG1) end interrupt	18	4	LPS
INT_SG2	Scan group 2 (SG2) end interrupt	19	5	LPS
INT_SG3	Scan group 3 (SG3) end interrupt	20, 32	6	LPS
ADC_CONV_END0	Scan group 4 (SG4) A/D conversion end signal	—	7	—
ADCA1				
INT_ADE	A/D error interrupt	212	—	—
INT_SG1	Scan group 1 (SG1) end interrupt	213	103	—
INT_SG2	Scan group 2 (SG2) end interrupt	214	104	—
INT_SG3	Scan group 3 (SG3) end interrupt	215	105	—
ADC_CONV_END1	Scan group 4 (SG4) A/D conversion end signal	—	106	—

31.1.5 Reset Sources

ADCAn reset sources are listed in the following table. ADCAn is initialized by these reset sources.

Table 31.9 Reset Sources

Unit Name	Reset Source
ADCA0	Reset sources other than transition to DeepSTOP mode (AWORES)
ADCA1	All reset sources (ISORES)

31.1.6 External Input/Output Signals

External input/output signals of ADCAn are listed below.

Table 31.10 ADCA0 Analog Input Signals

Unit Signal Name	Alternative Port Pin Signal	Resolution	T&H	RH850/F1K 100 pins	RH850/F1K 144 pins	RH850/F1K 176 pins
ANI000	ADCA0I0	12	√	√	√	√
ANI001	ADCA0I1	12	√	√	√	√
ANI002	ADCA0I2	12	√	√	√	√
ANI003	ADCA0I3	12	√	√	√	√
ANI004	ADCA0I4	12	√	√	√	√
ANI005	ADCA0I5	12	√	√	√	√
ANI006	ADCA0I6	12	—	√	√	√
ANI007	ADCA0I7	12	—	√	√	√
ANI008	ADCA0I8	12	—	√	√	√
ANI009	ADCA0I9	12	—	√	√	√
ANI010	ADCA0I10	12	—	√	√	√
ANI011	ADCA0I11	12	—	√	√	√
ANI012	ADCA0I12	12	—	√	√	√
ANI013	ADCA0I13	12	—	√	√	√
ANI014	ADCA0I14	12	—	√	√	√
ANI015	ADCA0I15	12	—	√	√	√
ANI016	ADCA0I0S	10	—	√	√	√
ANI017	ADCA0I1S	10	—	√	√	√
ANI018	ADCA0I2S	10	—	√	√	√
ANI019	ADCA0I3S	10	—	√	√	√
ANI020	ADCA0I4S	10	—	√	√	√
ANI021	ADCA0I5S	10	—	√	√	√
ANI022	ADCA0I6S	10	—	√	√	√
ANI023	ADCA0I7S	10	—	√	√	√
ANI024	ADCA0I8S	10	—	√	√	√
ANI025	ADCA0I9S	10	—	√	√	√
ANI026	ADCA0I10S	10	—	√	√	√
ANI027	ADCA0I11S	10	—	√	√	√
ANI028	ADCA0I12S	10	—	√	√	√
ANI029	ADCA0I13S	10	—	√	√	√
ANI030	ADCA0I14S	10	—	√	√	√
ANI031	ADCA0I15S	10	—	√	√	√
ANI032	ADCA0I16S	10	—	√	√	√
ANI033	ADCA0I17S	10	—	√	√	√
ANI034	ADCA0I18S	10	—	√	√	√
ANI035	ADCA0I19S	10	—	√	√	√

Table 31.11 ADCA0 External Input/Output Signals

Unit Signal Name	Description	Alternative Port Pin Signal
ADCA0		
ADCA0TRG0	External trigger pin (scan group 1)*1	ADCA0TRG0
ADCA0TRG1	External trigger pin (scan group 2)*1	ADCA0TRG1
ADCA0TRG2	External trigger pin (scan group 3)*1	ADCA0TRG2
ADCA0SEL0	External analog multiplexer (MPX) output pin 0	ADCA0SEL0
ADCA0SEL1	External analog multiplexer (MPX) output pin 1	ADCA0SEL1
ADCA0SEL2	External analog multiplexer (MPX) output pin 2	ADCA0SEL2

Note 1. When the external trigger pin is used, the noise filter for the port needs to be set. For details, see **Section 2.12, Noise Filter & Edge/Level Detector**.

CAUTION

When port P8_6 is used as ADCA0I8S, port P8_6 pin outputs a low-level $\overline{\text{RESETOUT}}$ signal while a reset is asserted and continues to output a low level after the reset is deasserted.
For details, see Section 2.11.1.1, P8_6: $\overline{\text{RESETOUT}}$.

Table 31.12 ADCA1 Analog Input Signals

Unit Signal Name	Alternative Port Pin Signal	Resolution	T&H	RH850/F1K 100 pins	RH850/F1K 144 pins	RH850/F1K 176 pins
ANI100	ADCA1I0	12	—	—	√	√
ANI101	ADCA1I1	12	—	—	√	√
ANI102	ADCA1I2	12	—	—	√	√
ANI103	ADCA1I3	12	—	—	√	√
ANI104	ADCA1I4	12	—	—	√	√
ANI105	ADCA1I5	12	—	—	√	√
ANI106	ADCA1I6	12	—	—	√	√
ANI107	ADCA1I7	12	—	—	√	√
ANI108	ADCA1I8	12	—	—	—	√
ANI109	ADCA1I9	12	—	—	—	√
ANI110	ADCA1I10	12	—	—	—	√
ANI111	ADCA1I11	12	—	—	—	√
ANI112	ADCA1I12	12	—	—	—	√
ANI113	ADCA1I13	12	—	—	—	√
ANI114	ADCA1I14	12	—	—	—	√
ANI115	ADCA1I15	12	—	—	—	√
ANI116	ADCA1I0S	10	—	—	√	√
ANI117	ADCA1I1S	10	—	—	√	√
ANI118	ADCA1I2S	10	—	—	√	√
ANI119	ADCA1I3S	10	—	—	√	√
ANI120	ADCA1I4S	10	—	—	—	√
ANI121	ADCA1I5S	10	—	—	—	√
ANI122	ADCA1I6S	10	—	—	—	√
ANI123	ADCA1I7S	10	—	—	—	√

Table 31.13 ADCA1 External Input/Output Signals

Unit Signal Name	Description	Alternative Port Pin Signal
ADCA1		
ADCA1TRG0	External trigger pin (scan group 1)*1	ADCA1TRG0
ADCA1TRG1	External trigger pin (scan group 2)*1	ADCA1TRG1
ADCA1TRG2	External trigger pin (scan group 3)*1	ADCA1TRG2

Note 1. When the external trigger pin is used, the noise filter for the port needs to be set. For details, see **Section 2.12, Noise Filter & Edge/Level Detector**.

31.2 Overview

31.2.1 Functional Overview

ADCA has the following features.

- 10-bit/12-bit resolution
- Successive approximation conversion method
- Number of A/D input channels
  A/D conversion is available for a maximum of 36 ADCA0 channels and 24 ADCA1 channels. Additionally, ADCA0 supports the connection of an external analog multiplexer (MPX) to extend the number of analog input channels.
- Internal track and hold (T&H) circuit
  ANI000 to ANI005 (ADCA0I0 to ADCA0I5) of ADCA0 include the track and hold circuit. The track and hold circuit can sample up to 6 channels of analog input simultaneously.
- A/D conversion control by scan groups
  The A/D conversion channel or conversion mode (scan mode) can be set for each scan group.
- Two scan modes
  Multi-cycle scan mode: Specified number of scans are executed.
  Continuous scan mode: Scans are executed repeatedly and continuously.
- Asynchronous/synchronous suspend and resume function
  A processing for a scan group can be interrupted to run the processing for another scan group.
- Start trigger for each scan group
  Software, hardware, and external trigger can start processing of each scan group.
- Scan end interrupt and DMA transfer are supported.
  For each scan group, an interrupt request to INTC can be issued or DMA transfer can be started, each time a processing for the virtual channel indicated by the end virtual channel pointer ends, or a virtual channel ends.
- A/D conversion channel repeat function
  A/D conversion is performed for the same channel two or four times sequentially, and the result is stored in the data register.
- Abundant safety functions
  Abundant safety functions are provided, such as A/D converter diagnosis, diagnosis of the channel multiplexer, diagnosis of open pins, diagnosis of the T&H circuit, upper limit/lower limit check for the A/D converter, overwrite check for data registers, and read and clear function for data registers.
- Shortest A/D conversion time per channel
  1.15 µs (when MPX is not used)
  2.30 µs (when MPX is used)